NOT MEASUREMENT SENSITIVE

MIL-PRF-32405 (MR)
06 January 2013

PERFORMANCE SPECIFICATION

CLEANER, HAND WIPE, FOR AVIATION and MISSILE SYSTEMS, METALLIC SUBSTRATES, LOW OR EXEMPT VOC

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand wipe cleaners that do not contain hazardous air pollutants (HAP) (Clean Air Act 112(b)) and contain a low content of volatile organic compounds (VOC) or an exempt VOC (40 CFR 51.100) for use on aviation and missile systems.

1.2 Classification. The cleaners are of the following classes, types, and grades (see 6.2).

1.2.1 Classes.

Class 1. Aqueous (contains water).

Class 2. Non-Aqueous, Non-Halogenated Solvents.

Class 3. Non-Aqueous, Halogenated Solvents

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Materials and Manufacturing Technology Branch, Specifications and Standards Office, Attn: RDRL-WMM-D, Aberdeen Proving Ground, MD 21005-5069 or emailed to richard.j.squillacioti.civ@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 6850

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1.2.2 Types.

Type I: VOC Free or VOC Exempt

Type II: HAP Free with Vapor Pressure of ≤ 1 mm Hg [at 68° F (20° C)]

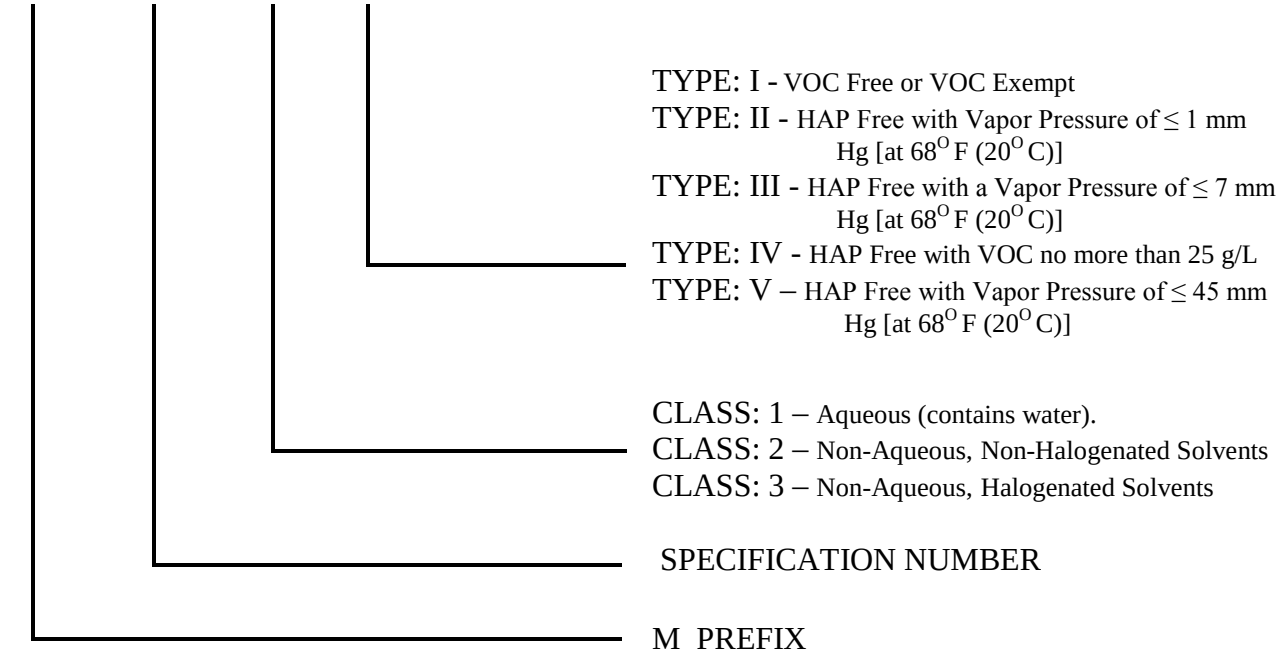
Type III: HAP Free with a Vapor Pressure of ≤ 7 mm Hg [at 68° F (20°C)]

Type IV: HAP Free with VOC no more than 25 g/L

Type V: HAP Free with Vapor Pressure of ≤ 45 mm Hg [at 68° F (20° C)]

1.3 Part or Identifying Number (PIN). PINs to be used for cleaners acquired by this specification are created as follows:

M 32405 1 II



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

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2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

A-A-58054	-	Abrasive Mats, Non-Woven, Non-Metallic

FEDERAL STANDARDS

FED-STD-141	-	Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing
FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-680	-	Degreasing Solvent
MIL-DTL-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-A-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-PRF-22750	-	Coating, Epoxy, High-Solids
MIL-PRF-23377	-	Primer Coatings: Epoxy, High-Solids
MIL-DTL-53039	-	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-DTL-64159	-	Camouflage Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant
MIL-PRF-81322	-	Grease, Aircraft, General Purpose, Wide Temperature Range
MIL-DTL-81381	-	Wire, Electric, Polyimide-Insulated, Copper or Copper Alloy
MIL-PRF-81733	-	Sealing and Coating Compound, Corrosion Inhibitive
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
MIL-PRF-85582	-	Primer Coatings: Epoxy, Waterborne

(Copies of these documents are available online at https://assist.dla.mil/quicksearch/ or https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 51.100	-	Air Emissions Reporting Requirements
40 CFR 63	-	National Emission Standards for Hazardous Air Pollutants for Source Categories

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40 CFR 261	-	Identification and Listing of Hazardous Waste
40 CFR 401.15	-	Protection of Environment (Toxic pollutants)

(Copies of this document are available online at http://www.gpoaccess.gov/cfr/ or from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated (see 6.2), the issue in effect on date of invitation for bids or request for proposal should apply.

ASTM INTERNATIONAL

ASTM D130	-	Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test
ASTM D471	-	Standard Test Method for Rubber Property—Effect of Liquids
ASTM D740	-	Standard Specification for Methyl Ethyl Ketone
ASTM D1193	-	Standard Specification for Reagent Water
ASTM D1353	-	Standard Test Method for Nonvolatile Matter in Volatile Solvents for Use in Paint, Varnish, Lacquer, and Related Products
ASTM D2240	-	Standard Test Method for Rubber Property Durometer Hardness
ASTM D2109	-	Standard Test Methods for Nonvolatile Matter in Halogenated Organic Solvents and Their Admixtures
ASTM D2251	-	Standard Test Method for Metal Corrosion by Halogenated Organic Solvents and Their Admixtures
ASTM D3167	-	Standard Test Method for Floating Roller Peel Resistance of Adhesives
ASTM D3359	-	Standard Test Methods for Measuring Adhesion by Tape Test
ASTM E70	-	Standard Test Method for pH of Aqueous Solutions with the Glass Electrode
ASTM F22	-	Hydrophobic Surface Films by the Water-Break Test
ASTM F483	-	Standard Test Method for Total Immersion Corrosion Test for Aircraft Maintenance Chemicals
ASTM F484	-	Standard Test Method for Stress Crazing of Acrylic Plastics in Contact with Liquid or Semi-liquid Compounds
ASTM F485	-	Standard Test Method for Effects of Cleaners on Unpainted Aircraft Surfaces
ASTM F502	-	Standard Test Method for Effects of Cleaning and Chemical Maintenance Materials on Painted Aircraft Surfaces
ASTM F519	-	Standard Test Method for Mechanical Hydrogen Embrittlement Evaluation of Plating Processes and Service Environments
ASTM F945	-	Standard Test Method for Stress-Corrosion of Titanium Alloys by Aircraft Engine Cleaning Materials
ASTM F1110	-	Standard Test Method for Sandwich Corrosion Test
ASTM F1111	-	Standard Test Method for Corrosion of Low-Embrittling Cadmium Plate by Aircraft Maintenance Chemicals

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ASTM G30	-	Standard Practice for Making and Using U-Bend Stress-Corrosion Test Specimens
ASTM G44	-	Standard Practice for Exposure of Metals and Alloys by Alternate Immersion in Neutral 3.5 % Sodium Chloride Solution

(Copies of these documents are available from http://www.astm.org or ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

AMS-H-6875	-	Heat Treatment of Steel Raw Materials
AMS-P-83310	-	Plastic Sheet, Polycarbonate, Transparent
AMS-QQ-A-250/4	-	Aluminum Alloy 2024, Plate and Sheet
AMS-QQ-A-250/12	-	Aluminum Alloy 7075, Plate and Sheet
AMS-S-8802	-	Sealing Compound, Temperature Resistant, Integral Fuel Tanks and Fuel Cell Cavities, High Adhesion
AMS-S-8840	-	Steel Sheet and Strip, Corrosion-Resistant, Precipitation Hardening (AM350 and AM335),
AMS 2629	-	Fluid, Jet Reference
AMS 2759	-	Heat Treatment of Steel Parts – General Requirements
AMS-M-3171	-	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on
AMS 3204	-	Rubber, Synthetic Low-Temperature Resistant 25-35
AMS 3209	-	Chloroprene (CR) Rubber Weather Resistant 65-75
AMS 4377	-	Magnesium Alloy, Sheet and Plate, 3.0Al - 1.0Zn - 0.20Mn (AZ31B-H24), Cold Rolled, Partially Annealed
AMS 4911	-	Titanium Alloy, Sheet, Strip, and Plate, 6Al - 4V, Annealed
AMS 5046	-	Carbon Steel, Sheet, Strip and Plate (SAE 1020 and 1025) Annealed
AMS 5629	-	Steel, Corrosion-Resistant, Bars, Wire, Forgings, Rings, and Extrusions, 13Cr - 8.0Ni - 2.2Mo - 1.1Al, Vacuum Induction Plus Consumable Electrode Melted Solution Heat Treated, Precipitation Hardenable
AMS 6359	-	Steel, Sheet, Strip, and Plate 0.80CR - 1.8NI - 0.25MO (0.38 - 0.43C) (SAE 4340)
AMS 6512	-	Steel, Maraging, Bars, Forgings, Tubing, and Rings 18Ni - 7.8Co - 4.9Mo - 0.40Ti - 0.10Al Consumable Electrode Vacuum Melted, Annealed
SAE ARP 1755	-	Effect of Cleaning Agents on Aircraft Engine Materials

(Copies of these documents are available from http://www.sae.org or SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. The cleaning compound furnished under this specification shall be a product which has been tested by a certified 3rd party testing facility approved by the qualifying activity and has passed the qualification tests specified herein and has been listed or approved for listing on the applicable Qualified Products Database (QPD) by the qualifying activity (see 3.1.1, 3.1.2, 4.2, 4.3, 4.4 and 6.3) before contract award. Any change in the chemical formulation, the material, the process, or the procedure in manufacturing the cleaning compound shall necessitate its being qualified again. Any change requires a complete requalification. The material supplied under contract shall be identical, within manufacturing tolerances, to the products receiving qualification. Any cleaning compound that does not conform to all the qualification tests specified herein shall be removed from the QPD.

3.1.1 Retention of qualification. To retain qualification approval of products listed on the Qualified Products Database (QPD), the manufacturer shall be required to verify by certification to the qualifying activity that its product(s) comply with the requirements of this specification. Unless otherwise specified by the qualifying activity (see 6.3), the time of periodic verification by certification shall be in two-year intervals from the date of original qualification (see 4.3). The certification action is initiated by the qualifying activity.

3.1.2 Qualifying activity. The activity responsible for the collection and storage of cleaning compounds certifications that are required to be furnished under this specification for qualification and conformance testing is the Specifications & Standards Office, Materials Manufacturing Technologies Branch, Material & Manufacturing Science Division, ATTN: RDRL-WMM-D, Army Research Laboratory located at Aberdeen Proving Ground, MD, 21005-5069.

3.2 Material. The materials used in the cleaning compound that are supplied under this specification shall be as specified herein. The materials not specified shall be selected by the contractor and shall be subjected to all the provisions of this specification.

3.2.1 Composition. The composition of the cleaning compound shall be the same as that of the sample furnished for qualification testing. Any change in the formulation of a qualified product shall require requalification. A document specifying the composition shall be forwarded to the Commander, US Army Public Health Command, 5158 Blackhawk Road, Aberdeen Proving Ground, MD 21010-5403.

3.2.2 Prohibited materials. The cleaning compound shall not contain any hazardous compounds as defined in 40 CFR 261, toxic pollutants in 40 CFR 401.15, nor hazardous air pollutants in 40 CFR 63. The cleaning compound shall not contain detectable amounts of any of the following: abrasives, chromates, cadmium, lead, mercury, phenols, cresols, chlorinated solvents (except SNAP approved), except where specified within this specification. The following materials are prohibited unless they are being used as an essential active ingredient in the cleaner: sodium chloride, urea, sodium sulfate, nitrites, nitrates, sucrose or any sugars.

3.2.2.1 Toxicity. The cleaning compound shall not contain any chemical listed by the current report of known carcinogens of the National Toxicology Program (NTP) (see 6.4.1). Any carcinogenic components in the cleaner in a concentration of 0.1 percent or greater by weight or volume will be regarded as sufficient for the presence of a carcinogen in the cleaner. The cleaners supplied under

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this specification shall have no adverse effect on human health (see 6.5) when used for their intended purpose (see 6.1). Army toxicity questions and/or a toxicity clearance request should be addressed to: Commander, US Army Public Health Command.

3.2.2.2 Hazardous Air Pollutants. The cleaner shall contain no chemicals listed as Hazardous Air Pollutants (Section 112 (B) HAP List) (see 6.4.2). Any HAP components in the cleaner in a concentration of 1.0 percent (%) or greater by weight (wt) or volume (vol) will be regarded as the presence of a HAP in the cleaner. The product containing less than 1 % of HAP shall be considered as a HAP free cleaner. For carcinogenic HAPs see 3.2.2.1.

3.2.2.3 Documentation. A certified letter from the U.S. Army Public Health Command (USAPHC), confirming the requirements of this section (3.2.2) shall be sent to the Specifications & Standards Office, Materials Manufacturing Technologies Branch, Material & Manufacturing Science Division, ATTN: RDRL-WMM-D, Army Research Laboratory located at Aberdeen Proving Ground, MD, 21005-5069, in accordance with paragraph 3.2.2.1 and 3.2.2.2.

3.3 Performance and physical requirements. The cleaning compound shall meet all the requirements listed in Table I when tested in accordance with Table III.

3.4 Cleaning method. The cleaning method shall be identified as hand wipe.

3.5 Storage/shelf life. Acceptable storage shelf life shall be verified by the accelerated storage stability test as defined in Table I.

3.6 Disposal. Where possible, these cleaners should be recycled or biodegraded. Disposal of used cleaner should be in accordance with local, state and Federal regulations. Care should be taken to avoid mixing used cleaner with other waste materials, especially those containing halogenated solvents.

3.7 Standard contaminant. The standard contaminant used in 4.9.16 section b, 4.9.17 section b and j, 4.9.18 section b, and in 4.9.19 section b shall be prepared by mixing two parts by weight of hydraulic fluid (MIL-PRF-83282 or equivalent), one part by weight lubricating grease (MIL-PRF-81322 or equivalent), and one-tenth part by weight of carbon black. This mixture shall be applied to panels, baked for two hours at 55°C (130°F), and cooled to room temperature. All cleaners must be qualified to the standard contaminant. If other contaminants are agreed upon the standard contaminant must also be inspected and reported as the baseline for performance data across all qualified cleaners.

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TABLE I. Performance and physical requirements.

PROPERTY	REQUIREMENTS --- PASS/FAIL CRITERIA		
	CLASS 1 – AQUEOUS	CLASS 2 – NON-AQUEOUS NON-HALOGENATED	CLASS 3 – NON-AQUEOUS HALOGENATED
Total Immersion Corrosion Test for Soak Tank Metal Cleaners	No staining, etching, pitting, no weight change exceeding Table II Limits	No staining, etching, pitting, no weight change exceeding Table II Limits	No staining, etching, pitting, no weight change exceeding Table II Limits
Corrosion of Low-Embrittling Cadmium Plate by Aircraft Maintenance Chemicals	No weight change > (0.14 mg/sq cm)/24 hrs	No Requirement	No Requirement
Sandwich Corrosion Test	No corrosion rating greater than reagent water Table II Alloys	No Requirement	No Requirement
Mechanical Hydrogen Embrittlement Evaluation of Plating Processes and Service Environments	No failures after 150 hours immersed. Type 1.d coupon	No Requirement	No Requirement
Effect of Cleaners on Unpainted Aircraft Surfaces	No streaking, staining not easily removed by hand pressure and water.	No streaking, staining not easily removed by hand pressure and water.	No streaking, staining not easily removed by hand pressure and water.
Stress Corrosion of Titanium Alloys by Aircraft Engine Cleaning Materials	No cracks in Table III Titanium Alloy	No Requirement	No cracks in Table III Titanium Alloy
Stress Corrosion of other alloys	No cracks in Table VI alloys	No cracks in Table VI alloys	No cracks in Table VI alloys
Detection of Copper Corrosion from Petroleum Products by the Copper Strip Tarnish Test	No Requirement	1b (Test for three hours at 100° C (212° F))	1b (Test for three hours at 100° C (212° F))
Effects of Cleaning and Chemical Maintenance Materials on Painted Aircraft Surfaces	No discoloration or decrease >1 pencil hardness	No discoloration or decrease >1 pencil hardness	No discoloration or decrease >1 pencil hardness
Stress Crazing of Acrylic Plastics in Contact with Liquid or Semi-Liquid Compounds	No crazing or staining	No crazing or staining	No crazing or staining
Stress Crazing of Polycarbonate Plastics	No crazing or staining	No crazing or staining	No crazing or staining
Effects on Polysulfide Sealants	No change ± 5 Shore A units	No change ± 5 Shore A units	No change ± 5 Shore A units
Effect on Polyimide Insulated Wire	No dissolution, cracking, or dielectric breakdown greater than wire exposed to de-ionized water	No dissolution, cracking, or dielectric breakdown greater than wire exposed to de-ionized water	No dissolution, cracking, or dielectric breakdown greater than wire exposed to de-ionized water

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TABLE I. Performance and physical requirements (Continued).

PROPERTY	REQUIREMENTS --- PASS/FAIL CRITERIA		
	CLASS 1 – AQUEOUS	CLASS 2 – NON-AQUEOUS NON-HALOGENATED	CLASS 3 – NON-AQUEOUS HALOGENATED
Rubber Property - Effect of Liquids	No detectable changes in mass or volume and no hardness changes greater than ±5 Shore A units shall occur after exposure to the test liquid.	No detectable changes in mass or volume and no hardness changes greater than ± 5 Shore A units shall occur after exposure to the test liquid.	No detectable changes in mass or volume and no hardness changes greater than ± 5 Shore A units shall occur after exposure to the test liquid.
pH Value	Less than pH 10 for cleaners mixed at the maximum supplier recommended concentration.	No Requirement	No Requirement
Nonvolatile Matter in Volatile Solvents Used in Paint, Varnish, Lacquer, and Related Products	No Requirement	<8.0 mg/100 mL;	<8.0 mg/100 mL;
Shelf-life	Specific shelf-life requirements should be specified in the contract or purchase order		
Accelerated Storage Stability	No separation or effect on steel or aluminum strips. Cleaning efficiency shall be no less than freshly made control formula.	No separation or effect on steel or aluminum strips. Cleaning efficiency shall be no less than freshly made control formula.	No separation or effect on steel or aluminum strips. Cleaning efficiency shall be no less than freshly made control formula.
Low Temperature Stability	No separation	No Requirement	No Requirement
Water Breakfree	Free of hydrophobic contaminants by this test method if the draining water layer remains as a thin continuous film over the surface for 1 min after withdrawal of the surface from the overflow container.	Free of hydrophobic contaminants by this test method if the draining water layer remains as a thin continuous film over the surface for 1 min after withdrawal of the surface from the overflow container.	Free of hydrophobic contaminants by this test method if the draining water layer remains as a thin continuous film over the surface for 1 min after withdrawal of the surface from the overflow container.
Sealant Adhesion	100% cohesive failure, minimum 3.5 kN/m (20 lbf/in)	100% cohesive failure, minimum 3.5 kN/m (20 lbf/in)	100% cohesive failure, minimum 3.5 kN/m (20 lbf/in)
Adhesive Bonding	Peel strength and % cohesive failure shall meet or exceed bond specimens cleaned with control cleaner	Peel strength and % cohesive failure shall meet or exceed bond specimens cleaned with control cleaner	Peel strength and % cohesive failure shall meet or exceed bond specimens cleaned with control cleaner
Paint Adhesion	No separation of primer to substrates.	No separation of primer to substrates.	No separation of primer to substrates.
Metal Corrosion by Halogenated Organic Solvents	No Requirement	No Requirement	No Corrosion

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TABLE II. Alloy and maximum average weight loss per grade. [1/] [2/]

Material [2/]	MAXIMUM WEIGHT LOSS (mg/cm^2/168 hours)	STOCK LOSS (µm/8 hrs.)	STOCK LOSS (mils/year)
AM-355 CRT	0.49	0.116	5
PH 13-8 Mo (H1000 condition, UTS approx. 205 KSI)	0.49	1.16	50
Maraging C-250 (aged 480° C/896° F for 3 hrs, approx. UTS 260 KSI)	0.49	1.16	50
Magnesium -- AZ31B-H24	0.7	1.16	50
Al 7075-T6 (Bare, UTS approx. 83 KSI)	0.49	1.16	50
Al 2024-T3 (Bare, UTS approx. 65 KSI)	0.49	1.16	50
Ti 6Al-4V	0.35	0.464	20
Steel 4340 (Oil quench from 800°C /1472° F, tempered 595°C/1100° F, UTS approx. 180-200 KSI)	0.49	1.16	50

[1/] Average of three panels. [2/] Material requirements per Table VII

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2)
b. Retention inspection (see 4.3)
c. Conformance inspection (see 4.4)

4.2 Qualification inspection. The qualification inspection shall consist of all tests specified in Table III. Failure of any test shall be cause for rejection. An infrared spectrum of the non-volatile matter for Class 1 Aqueous, and a spectrum of Class 2 Non-Aqueous and Class 3 Halogenated Solvents shall be recorded using an infrared spectrometer with a minimum resolution of 4 wavenumbers. Results shall be expressed as percentage transmittance in the range of 4,000 to 400 wavenumbers.

4.2.1 Qualification samples. The initial qualification samples shall consist of the required amount of the cleaning compound that is needed by the testing facilities (see 6.3.2). The cleaning compound shall be of the class, and type to be used in filling contract orders. Samples shall be identified (see 6.3.1.1) and forwarded to the appropriate testing facility, and the subsequent testing reports (see 4.2.2) shall be forwarded to the qualifying activity (see 3.1.2) for retention.

4.2.2 Test reports. The contractor shall provide certified test reports showing that the material conforms to all the requirements of this specification. The reports shall be submitted to the Specifications & Standards Office, Materials Manufacturing Technologies Branch, Material & Manufacturing Science Division, Attn: RDRL-WMM-D, Army Research Laboratory located at Aberdeen Proving Ground, MD, 21005-5069 (see 6.3.1). All test data described with the requirement of being noted, recorded, etc., in any section or table of this document is required to be reported.

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4.3 Retention inspection. To ensure compliance with all the requirements of this specification the manufacturer shall complete DD Form 1718 (Certification of Qualified Products) for each plant location and certify that the statements listed on the DD Form 1718 are true. Unless otherwise specified by the qualifying activity (see 6.2), the time of periodic verification by certification shall be in two-year intervals from the date of original qualification (see 3.1.1).

TABLE III. Properties and test methods.

TEST TYPE	TEST METHOD and/or REFERENCE PARAGRAPH	TESTING	
		QUALIFICATION	CONFORMANCE
Total Immersion Corrosion Test for Soak Tank Metal Cleaners	ASTM F483 in accordance with Paragraph.: 4.9.1	X	X
Corrosion of Low-Embrittling Cadmium Plate by Aircraft Maintenance Chemicals	ASTM F1111 in accordance with Paragraph.: 4.9.2	X	
Sandwich Corrosion Test	ASTM F1110 in accordance with Paragraph.: 4.9.3	X	
Mechanical Hydrogen Embrittlement Evaluation of Plating Processes and Service Environments	ASTM F519 in accordance with Paragraph.: 4.9.4	X	
Effect of Cleaners on Unpainted Aircraft Surfaces	ASTM F485 in accordance with Paragraph.: 4.9.5	X	
Stress Corrosion of Titanium Alloys by Aircraft Engine Cleaning Materials Table V	ASTM F945 Method A in accordance with Paragraph.: 4.9.6	X	
Stress Corrosion of other alloys Table VI	ASTM G30 in accordance with Paragraph: 4.9.6	X	
Detection of Copper Corrosion from Petroleum Products by the Copper Strip Tarnish Test	ASTM D130 in accordance with MIL-PRF-680	X	X
Effects of Cleaning and Chemical Maintenance Materials on Painted Aircraft Surfaces	ASTM F502 in accordance with Paragraph: 4.9.7	X	
Stress Crazing of Acrylic Plastics in Contact with Liquid or Semi-Liquid Compounds	ASTM F484 in accordance with Paragraph: 4.9.8	X	X
Stress Crazing of Polycarbonate Plastics	See Paragraph 4.9.9	X	
Effects on Polysulfide Sealants	See Paragraph 4.9.10	X	
Effect on Polyimide Insulated Wire	See Paragraph 4.9.11	X	
Rubber Property - Effect of Liquids	See Paragraph 4.9.12	X	X
pH Value	ASTM E70 Standard Test Method for pH of Aqueous Solutions With the Glass Electrode	X	X

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TABLE III. Properties and test methods. (Continued).

TEST TYPE	TEST METHOD and/or REFERENCE PARAGRAPH	TESTING	
		QUALIFICATION	CONFORMANCE
Nonvolatile Matter in Volatile Solvents Used in Paint, Varnish, Lacquer, and Related Products	ASTM D1353 in accordance with Paragraph: 4.9.13	X	
Accelerated Storage Stability	See Paragraph 4.9.14	X	
Shelf-life	See 6.2 & 6.6	Specific shelf-life requirements should be specified in the contract or purchase order	
Low Temperature Stability	See Paragraph 4.9.15	X	
Water Breakfree	ASTM F22 in accordance with Paragraph 4.9.16	X	X
Sealant Adhesion	See Paragraph 4.9.17	X	
Adhesive Bonding	See Paragraph 4.9.18	X	
Paint Adhesion	See Paragraph 4.9.19	X	
Metal Corrosion by Halogenated Organic Solvents	ASTM D2251	X	

4.4 Conformance inspection. Conformance inspection shall consist of the satisfaction of all of the requirements identified as conformance in Table III. If a lot fails any of the above acceptance tests during conformance testing, all required qualification tests shall be reinstituted. These qualification tests shall be required until two successive lots meet all requirements of the specification, after which conformance testing shall again be authorized.

4.5 Lot. A lot shall consist of one of the following.

4.5.1 Lot A. The cleaning compound produced in not more than 24 consecutive hours from a continuous process which is used to fill shipping containers directly from the process output. A continuous process shall be the production of product by continuous input of raw materials and output of finished product by one manufacturer in one plant with no change in manufacturing conditions or materials.

4.5.2 Lot B. The cleaning compound from individual runs of a batch process which is used to fill shipping containers directly from the process output. A batch process shall be the production of product by runs from single additions of raw materials which are mixed, reacted, or purified forming the product.

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4.5.3 Lot C. The cleaning compound from either or both the continuous and batch processes shall be held in a single storage tank and subsequently withdrawn to fill shipping containers. The product shall be homogeneous at the time of withdrawal and shall not be added to while being withdrawn. After each addition to the storage tank, the contents shall constitute a separate lot.

4.6 Sampling of product. Unless otherwise specified in the contract or purchase order (see 6.2), conformance inspection (4.4) shall be made on the sample of product taken directly from the filled containers. The number of filled containers selected for sampling from each lot shall be in accordance with Table IV. The first and last containers to be filled within a given lot shall be sampled. Other containers shall be selected at random. The samples may be obtained in any convenient manner that does not compromise the integrity of the sample.

4.7 Inspection of materials. The contractor is responsible for ensuring that materials and components used are manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification (see 2.4)

4.8 Rejection and retest. When any sample of the product examined and tested in accordance with this specification fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected. Rejected material shall not be resubmitted for acceptance without prior approval of the qualifying activity. The application for resubmission shall contain full particulars concerning previous rejections and all measures taken to correct those defects. Samples for retest shall be taken only from a sealed container.

TABLE IV. Number of Containers to be sampled.

Number of Containers in lot	Number of Containers to be sampled
2 to 15	2
16 to 25	3
26 to 90	5
91 to 150	8
151 to 280	13
281 to 500	20
501 to 1200	32
1201 to 3200	50
3201 to 10000	80
10001 to 35000	125
35001 to 150000	200
150001 to 500000	315
500001 and over	500

4.9 Testing methods. The tests of this specification shall be conducted in accordance with the applicable methods specified in Table III and listed in 4.9.1 to 4.9.19. The cleaning compound that is required to satisfy all the tests specified in 4.9 shall be selected at random from each lot. The contents of each selected container for sampling shall be thoroughly mixed by rolling and inverting immediately prior to sampling.

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4.9.1 Total immersion corrosion. The test procedure in ASTM F483 shall verify the cleaner's effect of total immersion corrosion. The alloys listed in Table II shall be used, except that the maximum intended use concentration of the cleaner shall be used when the procedure calls for testing at the recommended use concentration.

4.9.2 Low-embrittling cadmium plate corrosion. The cleaner's effect on low-embrittling cadmium plate corrosion shall be verified per ASTM F1111. The maximum intended use concentration of the cleaner shall be used when testing dictates use concentration.

4.9.3 Sandwich corrosion. The sandwich corrosion for alloys listed in Table II shall be verified per ASTM F1110. The maximum intended use concentration of the cleaner shall be used when testing dictates use concentration.

4.9.4 Hydrogen embrittlement. Hydrogen embrittlement of the cleaner shall be verified per ASTM F519, using Type 1d specimens.

4.9.5 Effects on unpainted surface. The alloys listed in Table II shall be verified per ASTM F485. The maximum intended use concentration of the cleaner shall be used when testing dictates use concentration.

4.9.6 Stress corrosion. The following procedures shall be used to verify the effects of the cleaner with regard to Stress Corrosion Cracking.

A. Procedures for Halogenated Solvent cleaners: (1) Using titanium alloy coupon in Table V, follow the procedure given in ASTM F945 with the following exceptions: Not every sheet of AMS 4911 is sensitive to stress corrosion cracking (SCC) and the sensitivity of the sheet should be verified per the test procedure in ASTM F945 prior to testing in the cleaner solutions (prior to purchasing if possible). Test a total of six specimens. Three specimens remain untreated after acid cleaning, and the other three specimens are tested in the candidate solvent. (2) Replace furnace heating Method A (900° F for 8 hours) or Method B (500° F for 168 hours), with the following: Place three test specimens in a 500 ml reaction flask fitted with multiple ports and closed with an o-ring seal. Add sufficient solvent to form a layer of solvent one-inch deep. Attach a reflux condenser, insert a thermometer or thermocouple, and heat over a light shielded 150 watt light bulb and reflux for 8 hours a day for 5 days. Record the temperature of the heated solvent during the testing. The bulb heating can be replaced by heating with a bath type apparatus where the bath temperature is controlled.

B. Procedures for Aqueous and Organic (non-halogenated) cleaners:

1. Effects on Titanium Alloys

   Follow the procedure given in ASTM F945 using Method A heat treatment. The coupon dimensions are provided in Table V. Note: Not every sheet of AMS 4911 is sensitive to stress corrosion cracking (SCC) and the sensitivity of the sheet should be verified per the test procedure in ASTM F945 prior to testing in the cleaner solutions (prior to purchasing if possible).

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2. Effects on Other Alloys:

a. Fabricate six U-bend specimens per ASTM G30 from the alloys, tempers, and dimensions listed in Table VI. The rolling direction will be parallel to the long axis of the coupon. Critical items include, but are not limited to:

(1) Alloys PH 13-8 Mo and Maraging C-250 are not available in finished thickness sheet or strip and may be machined from rod or bar stock. The coupons shall be machined from minimum size stock available with multiple passes to minimize distortion due to residual stresses. The machining procedure and surface finish shall follow paragraph 8.3 with special attention to the long edges.

(2) All coupons shall be in the finished temper, thickness, surface finish, and ground edge condition prior to bending.

(3) Coupons shall be cleaned by the following process:

i. Hand wash in 10% Alconox® solution and rinse thoroughly with reagent DI water and dry.

ii. Hand wash in acetone to remove ink markings and soluble contaminants.

iii. Hand wash in methyl ethyl ketone (MEK) to further remove soluble contaminants, markings, and dried organic material.

iv. Rinse with fresh anhydrous isopropyl alcohol (IPA) and dry.

v. Place in a desiccator pending utilization.

(4) The coupons shall be bent and assembled in a two stage method similar to Figure 5 in ASTM G30

i. The included angle after spring back shall be 65 ± 5 degrees. Coupons bent to less than 60 degrees shall be discarded.

ii. The coupon will be electrically insulated from the stainless bolt and nut. This shall be accomplished by using plastic shoulder washers between the coupon and the bolt.

iii. Upon assembly, the legs of the U-bend specimen will be brought together using a vise so that they are close to being parallel, but no further. The legs of the specimen are then brought to parallel using the nut and bolt. Once the coupon is in this position, the nut and bolt will not be relaxed.

b. Hold three unexposed U-bend Assemblies for each alloy in a clean protected location under standard laboratory conditions (65-75° F, and 30-70% relative humidity) to establish the validity of the baseline alloy samples. Evaluate these samples per paragraph d at the end of 30 days.

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c. Test three U-bend Assemblies for each alloy per ASTM G44 for 30 days at 24 hours per day with the following exceptions:

(1) Replace the 3.5% sodium chloride solution with the cleaning solution under test.

(2) Replace the one hour cycle with a two hour cycle that includes a 20 minute immersion in the cleaning solution and 100 minutes in the drying cycle.

(3) Test the solutions at the maximum use concentration and temperature recommended on the technical data sheet for the cleaner.

(4) Maintain the cleaning solution per the manufactures recommendations including, but not limited to; replenishing the lost liquid with deionized water, and/or diluted cleaning solution or undiluted non-aqueous solvent. The solution shall be tested at least weekly during the 30 day test period to insure the product is being properly maintained. The entire test solution may be replaced with new solution as required. All test results, adjustments, replenishments, and replacements shall be recorded.

(5) At the end of the test, clean the samples in deionized water, rinse in anhydrous isopropyl alcohol, and evaluate these samples per paragraph d.

d. Evaluate all of the restrained specimens visually using 20-diameters magnification. If cracks are found on the untreated (control) assemblies (b.), the test is void for the alloy that cracked. If cracks are found on candidate-solution-treated specimens (c.), this shall be cause for rejection of the candidate cleaning material.

TABLE V. Stress Corrosion Cracking Resistance Test Coupon Requirements for Titanium.

Material	Material Form	Specification and Temper	Approximate UTS (KSI)	Coupon Thickness (inch)	Bend Diameter (inch)
Ti 6Al-4V	Sheet / Strip	AMS 4911 Solution treated and aged.	160	0.050	0.45
Coupon size: 0.75” wide by 3”long with 0.28” diameter holes on 2” centers per ASTM F945.					

4.9.7 Effects on painted surfaces. The cleaner’s effect on painted surfaces shall be verified per ASTM F502 for all coating systems listed in Table VIII. The maximum intended use concentration of the cleaner shall be used when testing dictates use concentration. Aircraft green shall be considered a preferred color. Test panels shall be made from 2024-T3 per SAE AMS QQ-A-250/4 sheet and treated per the requirements found in Table VIII.

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TABLE VI. Stress Corrosion Cracking Resistance Test Coupon (1) Requirements for Other Alloys.

Material	Material Form	Specification and Temper	Approximate UTS (KSI)	Coupon Thickness (inch)	Bend Diameter (inch)
AM-355	Sheet / Strip	Strap Pack Material, Cold Rolled and Tempered (ref SAE-AMS-S-8840B	220 (180 yield)	0.040 ± 0.001 (2)	0.4 (2)
PH 13-8 Mo	Bar / Rod (3)	AMS 5629 H1000 condition	205	0.062 ± 0.003	0.5
Maraging C-250	Bar / Rod (3)	UNS K92890* heat treat per AMS 2759/3 Aged 480° C (900° F) for 3 hours *See AMS 6512G for the specific chemistry for UNS K92890	260	0.062 ± 0.003	0.75
Magnesium AZ31B-H24	Sheet / Strip	AMS 4377 surface treated in accordance with AMS-M-3171 Type III	42	0.062 ± 0.003	0.75
Al 2024	Sheet / Strip	AMS QQ-A-250/4 Temper T3	65	0.062 ± 0.003	0.5
Steel 4340	Sheet / Strip	AMS 6359 AMS-H-6875	160	0.062 ± 0.003	0.5

(1) Coupon size: 0.75” wide by 3”long with 0.28” diameter holes on 2” centers. The holes may be machined after bending to allow better alignment in the test assembly.
(2) AM-355 strip can be obtained from a rotorcraft OEM at available thicknesses. The bend diameter shall be increased with increasing thickness.
(3) Cut oversize coupons from bar or rod and grind to finish dimensions.

4.9.8 Effects on acrylic plastic. Stress crazing of acrylic plastic (Types A, B, and C) shall be verified by ASTM F484. The cleaner shall be tested at the maximum intended use concentration.

4.9.9 Effects on polycarbonate plastic. Stress crazing of polycarbonate plastic shall be verified by following the test procedure given in ASTM F484, with the following exceptions. The acrylic plastic shall be replaced with polycarbonate plastic conforming to AMS-P-83310 and shall be stressed for 30 ± 2 minutes to an outer fiber stress of 2000 psi. The cleaner shall be tested at the maximum intended use concentration.

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TABLE VII. Metal Coupon Material Requirements [1/]

Material	Material Form	Specification and Temper	Approximate UTS (KSI)
AM-355	Sheet / Strip	Strap Pack Material, Cold Rolled and Tempered (ref AMS-S-8840)	220 (180 yield)
PH 13-8 Mo	Bar / Rod [2/]	AMS 5629 and H1000 condition	205
Maraging C-250	Bar / Rod [2/]	UNS K92890* and heat treat per AMS 2759 / 3 Aged 480 °C (900 °F) for 3 hours *See AMS 6512G for the specific chemistry for UNS K92890	260
Magnesium AZ31B-H24	Sheet / Strip	AMS 4377 surface treated in accordance with AMS-M-3171 Type III	42
Al 7075	Sheet / Strip	AMS QQ-A-250/12 and Temper T6	83
Al 2024	Sheet / Strip	AMS QQ-A-250/4 and Temper T3	65
Ti 6Al-4V	Sheet / Strip	AMS 4911 Solution treated and aged.	160
Steel 4340	Sheet / Strip	AMS 6359 / AMS-H-6875	160

[1/] If the material is specified for a test then this table defines the requirements for the material.
[2/] Cut oversize coupons from bar or rod and grind to finish dimensions

TABLE VIII. Coating Systems for test by Effects on Painted Surfaces ASTM F502 & Adhesion Tape Test per ASTM D3359

COATINGS TO BE TESTED		SURFACE TREATMENT
PRIMER	TOPCOAT	
MIL-PRF-23377 Class C	MIL-DTL-53039	MIL-A-8625 Type I Anodize
MIL-PRF-23377 Class N	NONE	MIL-A-8625 Type I Anodize
MIL-PRF-23377 Class N	MIL-DTL-53039	MIL-DTL-5541 Type I Conversion Coat
MIL-PRF-23377 Class C	NONE	MIL-DTL-5541 Type I Conversion Coat
MIL-PRF-23377 Class N	NONE	MIL-DTL-5541 Type II (TCP)
MIL-PRF-23377 Class C	MIL-DTL-53039	MIL-DTL-5541 Type II (TCP)
MIL-PRF-23377 Class N	MIL-PRF-22750	MIL-DTL-5541 Type II (TCP)
MIL-PRF-85582 Class C	MIL-DTL-53039	MIL-A-8625 Type I Anodize
MIL-PRF-85582 Class N	MIL-DTL-64159	MIL-DTL-5541 Type II (TCP)
MIL-PRF-85582 Class N	NONE	MIL-DTL-5541 Type I Conversion Coat

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4.9.10 Effect on polysulfide sealants. The cleaner's effect on polysulfide sealants shall be verified by following the procedure described below:

a. Preparation of test specimens. Mix sealants MIL-PRF-81733, Class 1, Grade B Type II; and AMS-S-8802 Class B, as specified by their respective manufacturers and press into a 1/8 inch thick sheet mold. Use the standard cure of 14 days at 25 ±3° C at 50% humidity, or 48 hours at 60 ±3° C. This will be the sheet stock for each sealant, and the specimens shall be cut from the sheet stock.

b. Test procedure. Measure the Shore A hardness of each sealant per ASTM D2240 before exposing them to cleaners. Immerse two specimens of each sealant in the cleaner at the maximum intended use concentration at room temperature for 30 minutes. Wipe off the cleaner, and test within 30 minutes for Shore A hardness.

4.9.11 Effects on polyimide-insulated wire. The procedure specified in this paragraph shall verify the effects of the cleaning solvent on polyimide insulated wire. Coil two segments of MIL-DTL-81381/11-20 wire approximately 61 cm (24 in) and place into separate 118 ml (4 oz) wide mouth jars. To one jar add sufficient amount of the concentrated cleaner to completely cover the wire coil. To the other jar (control sample) add sufficient de-ionized water (ASTM D1193, Type IV) to cover the wire coil. Cap both jars and store at room temperature (20 - 25° C) for 14 days. At the end of the storage period remove both coils, rinse thoroughly with de-ionized water and suspend to allow complete draining and drying. Uncoil the wires, examine each closely for dissolution, and report the results. The wire immersed in the cleaner shall perform as well as the wire immersed in de-ionized water. Both wires shall then be subjected to a double reverse wrap on a 0.3 cm (0.125 in) mandrel and examined for cracking. (Note: failure of the control sample here voids the test and must be repeated using new MIL-DTL-81381/11-20 material). Wire immersed in the cleaner shall then be examined for cracking. If cracking occurs results shall be reported and the test ended. Passing wire shall then withstand a one-minute dielectric test of 2,500 volts (rms). Using a Hypot model number 4045 or equivalent, examine for breakdown and/or leakage. Wire immersed in the cleaner shall perform equally well as the control wire immersed in de-ionized water.

4.9.12 Rubber Property - Effect of Liquids. Two representative elastomers: a) Chloroprene ® Rubber, Weather Resistant 65 – 75 (per SAE AMS 3209L) and b) Rubber Synthetic Low Temperature Resistant (per SAE AMS 3204M) shall have all sample surfaces immersion exposed to the cleaning fluid at 23° C ± 2° C (73° F ± 4° F) for 22 Hours. ASTM D471 governs the exposure conditions and property change analysis. Evaluation of change in mass, change in volume, and mass of soluble matter extracted by the liquid will be reported. The maximum intended use concentration shall be used. Additionally, changes in Durometer Hardness between non-exposed and exposed specimens will be investigated and reported, per ASTM D2240. The pass or fail criteria shall be that no detectable changes in mass or volume and no hardness changes greater than ± 5 Shore A Units shall occur after exposure to the test liquid.

4.9.13 Non-volatile residue. Class 3 cleaners' non-volatile residue shall be verified per ASTM D2109. Class 2 cleaners non-volatile residue shall be verified per ASTM D1353, with the following exception: For cleaners that may polymerize or react at 100° C, e.g., cleaners containing d-limonene or terpenes, the cleaner may be dried in a vacuum chamber at room temperature to evaporate the cleaner, instead of using a 100°C steam bath as called out by ASTM D1353.

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4.9.14 Accelerated Storage Stability.

4.9.14.1 Preparation of test sample. Two 20.4 oz (600 mL) portions of a well-shaken cleaning compound shall be poured into a clean pressure resistant clear glass bottles that are of sufficient size to contain the entire cleaner and to allow insertion of steel and aluminum coupons. A strip of steel 6.0 x 0.50 x 0.20 in (15.2 x 1.30 x 0.5 cm) conforming to SAE-AMS 5046 and a strip of 2024 aluminum conforming to AMS QQ-A-250/4 and Temper T3 of the same dimensions shall be polished with 280-grit silicon carbide paper to remove surface contamination and then cleaned by boiling for one minute in mineral spirits and one minute in isopropyl alcohol. The coupons shall be placed in the test bottles and the bottles capped. The bottles shall then be inverted to completely coat the coupons.

4.9.14.2 Procedure. The capped bottles containing the coupons shall be placed in a hot water bath held at 60° C ±2° C (140°F ± 4° F) for 8 hours. The bath and the bottles shall then be allowed to cool for 16 hours and the 24 hour cycle repeated each day for a total of seven days. On the eighth day, the bottles shall be removed from the bath, uncapped, examined for separation, and the coupons withdrawn and examined for corrosion. The aged samples shall be tested for water break in accordance with ASTM F22.

4.9.15 Low temperature stability. Pour approximately 50 ml of the concentrated, as-received cleaner into a test tube of suitable size and cool to 0 ± 2° F in a cold box. Keep the sample at this temperature for one hour. Remove the sample from the cold box and allow the temperature to rise to room temperature. Invert the test tube five times and examine the content for homogeneity.

4.9.16 Water Breakfree. The procedure described below (a – c) is carried out in order to prepare panels to be tested by the water break procedure, ASTM F22.

a) Test panels shall be 1 x 70 x 150 mm (0.04 x 2.75 x 6 inches) made from 2024-T3 per SAE AMS QQ-A-250/4 sheet and conversion coated per MIL-DTL-5541, Type I, Class 1A.

b) Apply the standardized contaminant mixture (paragraph 3.7), or other contaminants appropriate to the specific application, to three (3) test panels. Wipe the panel surfaces with a cleaning cloth to remove gross contaminants.

c) Wipe each test panel five times with a clean cloth soaked with the candidate cleaner.

4.9.17 Sealant adhesion. Sealant adhesion shall be verified by following the peel test procedure described below (a – j). Test panels shall be 1 x 70 x 150 mm (0.04 x 2.75 x 6 inches) made from 2024-T3 per SAE AMS QQ-A-250/4 sheet and conversion coated per MIL-DTL-5541, Type I, Class 1A. There shall be a minimum of eight test panels tested per primer.

a. Apply each of the primers listed in Table IX to the properly prepared test panels per the primer specification. Scuff sand the primed panels with abrasive mats conforming to A-A-58054 Type I Class 1 Grade A or equivalent. For each primer, follow the procedure described below.

b. Apply the standardized contaminant mixture (paragraph 3.7), or other contaminants appropriate to the specific application, to eight (8) test panels. Wipe the panel surfaces with a cleaning cloth to remove gross contaminants.

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c. Wipe four (4) test panels five times with a clean cloth soaked with the candidate cleaner. Wipe the other four (4) test panels with methyl ethyl ketone (ASTM D740) – the control cleaner. For Type 1 cleaner, use the cleaner at the intended use concentration and wipe panel surfaces again with a cloth soaked with de-ionized water (ASTM D1193, Type IV).

d. Coat at least 125 mm (5 inches) of the panels on one side with a 3 ± ½ mm ($^{1}/_{8} \pm {}^{1}/_{64}$ inch) thickness of sealing compound per AMS-S-8802 Class B.

e. Impregnate a 70 x 300 mm (2¾ x 12 inch) strip of wire screen (20 to 40 mesh aluminum or Monel wire fabric), or equivalent with sealant, such that approximately 125 mm (5 inches) of one end is completely covered with sealant on both sides. Work the sealant well into the fabric.

f. Place the sealant-impregnated end of the fabric on the sealant-coated panel. Smooth the fabric down on the panel, taking care not to trap air beneath the fabric.

g. Apply an additional 1mm ($^{1}/_{32}$ inch) thick coating of sealing compound over the fabric. Use the standard cure of 14 days at 25 ±2° C at 50% humidity, or 48 hours at 60 ±2° C.

h. Completely immerse the panels in jet reference fluid per AMS 2629 Type I at 60 ± 1°C (140 ± 2°F) for 7 days, using covered glass vessels.

i. Cool the panels in the jet reference fluid for 24 hours.

j. Measure the sealant peel strength within 10 minutes after removal from the jet reference fluid: Cut two 25-mm (1-inch) wide sections lengthwise through the fabric and sealing compound on each panel. In a suitable tensile testing machine, strip back the fabric at an angle of 180 degrees to the metal panel, using a rate of 50 mm/min (2 inches/min). During the peel strength testing, make three cuts at approximately 25-mm (1-inch) intervals through the sealing compound to the panel to attempt to promote adhesive failure. Measure the numerical average of the peak loads. Do not include failures of the sealing compound to the fabric in the peel strength values. Measure the percentage of cohesive/adhesive failures. If the control specimens that are prepared with methyl ethyl ketone do not meet the performance requirements, repeat the procedure with a different batch of sealant. Report the test results and the contaminants used, if other than the standard contaminant (paragraph 3.7)

TABLE IX. Primers for Sealant Adhesion and Paint Adhesion

Specification	Description
MIL-DTL-23377 class C1 or C2	Performance Specification Primer Coatings: Epoxy, High-Solids: CHROMATED
MIL-DTL-23377 class N	Performance Specification Primer Coatings: Epoxy, High-Solids: NON-CHROMATED
MIL-DTL-85582 class C1 or C2	Performance Specification Primer Coatings: Epoxy, Waterborne: CHROMATED
MIL-DTL-85582 class N	Performance Specification Primer Coatings: Epoxy, Waterborne: NON-CHROMATED

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4.9.18 Adhesive bonding. Adhesive bonding shall be verified by following the procedure described below (a – h) using aluminum (7075-T6 bare), stainless steel (AM-355), titanium (6Al-4V), and nickel (electroformed) substrates. The bonding surfaces of the test panels shall be roughened by mechanical abrading.

a. Cut a sufficient number of test panels of each substrate to provide a minimum of six specimens per cleaner for each of the two testing temperatures (i.e. minimum twelve test specimens per cleaner per substrate). Use panel thickness and dimensions that are appropriate to fabricate test specimens per ASTM D3167. The flexible adherends shall be 0.63 mm (0.025 in.) and the rigid adherends shall be 1.63 mm (0.064 in.).

b. Prepare and apply the standard contaminant according to paragraph 3.7.

c. Remove the gross contaminant with a clean cloth.

d. Hand wipe the candidate-cleaner panel surfaces until visually clean with a clean cloth soaked with the candidate cleaner at the specified use concentration. For aqueous cleaners, hand wipe panels again with de-ionized water (ASTM D1193, Type IV) until visual evidence of the cleaner is removed. Hand wipe the control-cleaner panel surfaces until visually clean with a clean cloth soaked with methyl ethyl ketone (ASTM D740).

e. Abrade the panel bonding surfaces with an orbital sander: 180 grit for aluminum, 120 grit for titanium, and 80 grit for stainless steel and nickel. Clean again per step d.

f. Fabricate test specimens per ASTM D3167, using a paste adhesive (Dexter Hysol EA 9309.3NA or equivalent). Use shims or scrim cloth, if necessary, to control the bondline thickness to 5 – 10 mils. Cure the adhesive as recommended by its manufacturer. Record the bondline thickness and the adhesive cure conditions.

g. Perform floating roller peel tests in accordance with ASTM D3167 on a suitable testing machine (e.g., Instron or equivalent). Condition and test at least six (6) specimens for each testing temperature as follows: (1) Condition at 25° C (77° F) / ambient humidity for 24 hours minimum. Test at 25°C (77° F). (2) Condition at 80° C (176° F) / 95% humidity environment for 30 days. Test at 80° C (176° F).

h. Examine/compare and record failure surface characteristics, particularly adhesive/ cohesive failure modes. Record measured peel strength values and any pertinent observations or anomalies. Record the contaminants used, if other than the standard contaminant, and the cleaner concentration used (for aqueous cleaner).

4.9.19 Paint adhesion. Paint adhesion shall be verified by following the procedure described below (a – e):

a. Prepare 3 x 100 x 100 mm (0.125 x 4 x 4 inch) aluminum test panels made from 2024-T3 per SAE AMS QQ-A-250/4 sheet and conversion coated per MIL-DTL-5541, Type I, Class 1A. There shall be a minimum of three test panels tested per primer. Allow panels to dry for 24 hours at room temperature.

b. Apply the standardized contaminant mixture (paragraph 3.7), or other contaminants appropriate to the specific application, to the test panels.

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c. Clean the panels with the candidate cleaner using the cleaning process that is recommended for the particular cleaner/application. Allow panels to dry at room temperature for 24 hours.

d. Apply each of the primers listed in Table IX to the properly prepared (soiled and cleaned) test panels per the primer specification. The primer coatings shall be allowed to air dry for not less than 14 days, or air dry for not less than one hour followed by 24 hours at 65.5 ± 3°C (150 ± 5° F) prior to testing. Then immerse the test panels in de-ionized water for 24 hours.

e. Test paint adhesion per FED-STD-141, Test Method No. 6301. Examine for any separation of primer from the substrates. Report the contaminants used, if other than the standard contaminant, and the cleaning process used, including the cleaner concentration, processing time, and temperature.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel components are to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cleaning compounds covered by this specification are intended to be used for hand wipe cleaning on metallic substrates prior to painting, metal surface treatment, non-structural sealant application and adhesive bonding. This specification applies to aerospace and missile end items/components and equipment as defined by National Emission Standards for Hazardous Air Pollutants (NESHAPs) for Aerospace Manufacturing and Rework Facilities and Defense Land Systems and Miscellaneous Equipment (DLSME). Cleaners approved to this specification are not intended for cleaning prior to die-penetrant inspection (FPI) unless specifically approved for FPI by the responsible technical authority. These cleaners should only be used when documented in approved DOD technical publications for maintenance of DOD materiel.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Specify class, type, grade, and category of the cleaner (see 1.2).
c. If issues of documents are different (see 2.2.1, 2.2.2, and 2.3).
d. If the time of periodic verification is different (see 3.1.1 and 4.3).
e. Specify shelf life requirements (see Table I, Table III, and 6.6).
f. If conformance tests are to be made on different samples then those specified (see 4.4.3).
g. If packaging requirements are different (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products Database QPD-32405, whether or not such products have actually been so listed by that date. The attention of manufacturers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information (questions, etc.) pertaining to qualification and testing of products and the certified test results from the approved testing facility for the qualified products database (QPD) testing (see 4.2) and for the conformance program (see 4.4) should be submitted to the Specifications & Standards Office, Materials Manufacturing Technologies Branch, Material & Manufacturing Science Division, Attn: RDRL-WMM-D, Army Research Laboratory located at Aberdeen Proving Ground, MD, 21005-5069.

6.3.1 Inspection and other information. In addition to the qualification test samples, the qualifying activity will request the manufacturer to submit to the qualification activity: (a) a certified test report showing that the material conforms to the requirements of this specification (see 4.2.2); (b) one copy of the MSDS (see 6.5.1); (c) certification that the manufacturer's material meets the HAP and VOC content requirement (see Table I, 1.2.2 and 3.2.2).

6.3.1.1 Qualification inspection sample identification. Qualification inspection samples are to be forwarded to the laboratory designated in the letter of authorization (see 6.3) and identified as follows:

QUALIFICATION TEST SAMPLES

Specification MIL-PRF-32405, Class 1, 2, or 3, Type I, II, III, IV, or V (as applicable)

CLEANER, HAND WIPE, FOR AVIATION and MISSILE SYSTEMS, METALLIC SUBSTRATES, LOW OR EXEMPT VOC

Manufacturer's name and product number
Submitted by (name and date) for qualification testing in accordance with authorization (reference authorizing letter)

6.3.2 Testing facilities. The testing facilities identified below can each do some of the required tests, however, all the tests can be performed by a combination of those facilities listed below. Please note that this list is not an exclusive list. The list was prepared for your convenience only. There are other facilities that are capable of performing the required tests. The South Coast Air Quality Management District (SCAQMD) can perform only the combined VOC Content and Vapor Pressure tests.

SMI, Inc.
12219 SW 131 Avenue
Miami, FL 33186-6401 USA
Ph: 305-971-7047
Attn: Patricia D. Viani, President

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Petro-Lube Test Labs.
116 Sunset Inn Road
P.O. Box 300
Lafayette, NJ 07848
Ph: 973-579-3448
Attn: Josiah Wintermute

South Coast Air Quality Management District
21865 Copley Dr.
Diamond Bar, CA 91765
Ph: 909-396-2000

Concurrent Technologies Corporation
7995 114th Avenue
Largo, FL 33773-5026
Michael Miller
Phone: 727-549-7285
Email: millermi@ctc.com

6.4 Definitions.

6.4.1 Carcinogens. The Occupational Safety and Health Administration (OSHA) definition of carcinogens is those chemicals / processes appearing in lists 1, 2A, and 2B of the International Agency for Research on Cancer (IARC) 1/; substances known to be carcinogenic and occupational exposures associated with a technological process known to be carcinogenic by the National Toxicology Program (NTP) Report on Carcinogens (latest annual report) 2/; and OSHA regulated carcinogens 3/.

1/ Copies of these volumes may be found in medical libraries or through the World Health Organization, 1211 Geneva 27, Switzerland.

2/ Copies may be obtained from the Public Health Service, National Toxicology Program, Public Information Office, P.O. Box 12233, MD B2-04, Research Triangle Park, NC 27709 or at http://ntp-server.niehs.nih.gov.

3/ Copies may be found in 29 CFR part 1910 Subpart Z.

6.4.2 Hazardous Air Pollutant (HAP). HAP is defined as any substance listed under Section 112 of the Clean Air Act or its modifications. The text of the Clean Air Act, listed pollutants and modifications are kept by the Environmental Protection Agency (EPA) and are accessible through the website: http://www.epa.gov.

6.4.3 Volatile organic compound (VOC). VOC is defined in 40 CFR Part 51.100. Volatile organic compounds (VOC) means any compound of carbon, excluding carbon monoxide, carbon dioxide,

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carbonic acid, metallic carbides or carbonates, and ammonium carbonate, which participates in atmospheric photochemical reactions. For the latest complete definition of VOC see: http://www.epa.gov/ttn/naaqs/ozone/ozonetech/def_voc.htm.

6.4.3.1 Exempt VOC. A list of exempt VOC is listed in 40 CFR Part 51.100 (s). For the current list of exempt organic compounds see: http://www.epa.gov/ttn/naaqs/ozone/ozonetech/def_voc.htm. Compounds that are explicitly exempted from regulation by the EPA as VOCs are those compounds of carbon that have negligible photochemical reactivity (they do not produce ozone in atmospheric reactions).

6.5 Material Safety Data Sheet (MSDS). An MSDS must be prepared and submitted in accordance with FED-STD-313. The MSDS must also meet the requirements of 29 CFR 1910.1200. The 29 CFR 1910.1200 requires that the MSDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Questions pertinent to the effect(s) of these cleaners on the health of personnel using them should be referred by the procuring activity to the appropriate medical service, who will act as its adviser. Contracting officers will identify the activities requiring copies of the MSDS.

6.6 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, Shelf-life Management Manual. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order:

(1) the Inventory Control Points that manage the item and
(2) the DoD Service and Agency administrators for the DoD Shelf-Life Program.

Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.hq.dla.mil/

6.7 Subject term (key word) listing.

Aqueous
Environmental
Non-aqueous
Non-hazardous
Support vehicles

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
ARMY – MR
(Project 6850-2011-005)

Review activities:
Army - AT, AV, EA, GL, MI
DLA - GS, GS3
GSA - FAS
MISC – MP

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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